Evaluation Report — Provider Comparison & Prompt Evaluation

Task: Email Classification Prompt Evaluation

Providers: Google Gemini and Groq (Llama 3.3 70B Versatile)

Test set: 10 email classification inputs

1. Objective

The objective was to evaluate the same classification prompt across two AI providers using the same 10 test inputs. Each model's output was compared with the expected category and marked Pass or Fail.

2. Test Results

#	Expected	Groq	Result	Gemini	Result
1	Sales	Sales	PASS	Sales	PASS
2	Support	Support	PASS	Support	PASS
3	Complaint	Complaint	PASS	Complaint	PASS
4	Invoice	Invoice	PASS	Invoice	PASS
5	Spam	Spam	PASS	Spam	PASS
6	General	General	PASS	General	PASS
7	Sales	Sales	PASS	Sales	PASS
8	Support	Support	PASS	Support	PASS
9	Complaint	Complaint	PASS	Complaint	PASS
10	Invoice	Support	FAIL	Invoice	PASS

3. Accuracy

Groq: 9/10 correct = **90%** accuracy.
Gemini: 10/10 correct = **100%** accuracy.

Groq's only error occurred on Test 10: the expected category was Invoice, but Groq returned Support.

4. Latency

Exact latency measurements were not recorded during the manual web-interface tests. Therefore, no numerical average latency is claimed. This is a limitation because web-interface response time can be affected by browser, network, and service conditions.

5. Estimated Cost

The tests were performed through web interfaces, so no actual API charge was measured. For an illustrative paid-API comparison, assume 100 input tokens and 10 output tokens per classification run.

Provider / Model	Input / 1M	Output / 1M	Illustrative / 1,000 runs
Groq — Llama 3.3 70B	$0.59	$0.79	$0.0669
Gemini 2.5 Flash	$0.30	$2.50	$0.0550

These are illustrative estimates, not actual charges from the 10 web-interface tests. Actual cost depends on token usage, model, and pricing tier.

6. Recommendation

Recommended provider: Gemini. Gemini achieved 100% accuracy compared with Groq's 90% on this test set. Under the illustrative token assumption, Gemini 2.5 Flash also has a slightly lower estimated API cost per 1,000 runs. Because only 10 inputs were tested and latency was not measured, a larger controlled test would be needed before making a production decision.

7. Conclusion

The evaluation showed that both providers performed well on the email-classification task, but Gemini produced the best accuracy on the selected test set. The experiment demonstrates the importance of measuring model performance with a consistent test set instead of choosing a provider based only on assumptions.